Exploratory Data Analysis of the Opioid- and Stimulant-related Harms in Canada Using R

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Final Project Milestone-1
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1. **Introduction**:

Canada is facing one of the most devastating public health crises in its modern history. Since 2016, tens of thousands of Canadians have died from opioid toxicity, with the situation worsening dramatically during the COVID-19 pandemic. Accidental poisonings are now the leading cause of death for males aged 20 to 49 and females aged 20 to 39 in Canada — a statistic that underscores both the scale and the demographic concentration of this emergency.

This report presents an Exploratory Data Analysis (EDA) of the Opioid- and Stimulant-related Harms in Canada dataset, published quarterly by the Public Health Agency of Canada (PHAC). The dataset was downloaded directly from the PHAC Public Health Infobase and was last updated in March 2026, covering January 2016 through September 2025. This is raw government surveillance data that has not been pre-analyzed in any machine learning repository, making it well-suited for independent exploratory analysis.

This dataset was selected because it is current, nationally significant, and directly relevant to ongoing public health policy decisions in Canada. Unlike static historical datasets, this data is updated every quarter — meaning the findings of this analysis speak to a living crisis. The analysis addresses the following questions:

- How have opioid and stimulant deaths trended nationally from 2016 to 2024, and what role did COVID-19 play?
- Which provinces carry the highest burden of opioid-related deaths and what does the geographic distribution reveal?
- Which demographic groups — by age and sex — are most affected?
- How has the role of fentanyl in opioid deaths changed over time?
- Are all four harm types (deaths, hospitalizations, ED visits, EMS responses) trending together?

1.1 **Dataset Description:**

The dataset contains 24,464 rows and 11 columns. Each row represents a single data point for a specific combination of substance, harm type, geographic region, time period, and demographic breakdown. Table 1 describes each field.

Table 1: Data Dictionary — PHAC Opioid- and Stimulant-related Harms Dataset

Field	Type	Description
Substance	Categorical	Opioids or Stimulants
Source	Categorical	Deaths, Hospitalizations, ED Visits, or EMS Responses
Specific_Measure	Categorical	Dimension of breakdown: Overall numbers, Age group, Sex, Type of opioids, etc.
Region	String	Canada, province, territory, or sub-regional area (19 unique regions)
PRUID	Integer	Province/Territory unique identifier (Statistics Canada standard)

Time_Period	Categorical	"By year" (annual) or "By quarter" (quarterly)
Year_Quarter	String	Specific year (e.g., "2024") or quarter (e.g., "2023 Q2")
Aggregator	String	Year or age-group context — blank for overall totals (27% missing)
Disaggregator	String	Sub-category (e.g., "Male", "Fentanyl", "30 to 39 years") — blank for overall totals (27% missing)
Unit	Categorical	Number (count), Crude rate (per 100,000), or Percent
Value	String	Data value — numeric for valid records; "Suppr." for suppressed small counts (9.6%)

1.2 **Data Cleaning:**

Prior to analysis, four cleaning steps were applied to the raw dataset. No rows were deleted — all 24,464 original records were retained.

Step 1 — Blank to NA conversion: The Aggregator and Disaggregator columns contained empty strings ("") for 6,596 rows (27.0%) representing overall national or provincial totals not broken down by any demographic sub-category. These blank strings were converted to proper NA values so that R's is.na() function could correctly filter these rows. This is by design in the original data — not an error — and is the essential structural feature of the dataset that must be understood before any filtering can work correctly.

Step 2 — Numeric conversion: The Value column is stored as a string because it contains both numeric values and the text "Suppr." (meaning suppressed). A new numeric column Value_num was created using suppressWarnings(as.numeric()), converting valid numeric strings to numbers and non-numeric entries to NA. This created 8,105 NAs — 2,357 from suppressed values and the remainder from rows containing non-numeric formatting.

Step 3 — Suppressed value flagging: A Boolean column Suppressed was created to explicitly flag the 2,357 rows (9.6%) containing "Suppr." in the Value field. These values are suppressed by PHAC and provincial coroners when counts are very small — typically fewer than 5 events — to protect the privacy of individuals. These rows are retained in the dataset with a flag so they can be excluded from quantitative analysis while remaining visible for completeness.

Step 4 — Year extraction: A clean 4-digit numeric Year_clean column was extracted from Year_Quarter using a regular expression. This correctly handles complex entries such as "2025 (Jan to Sep)", extracting only the year digit for sorting and filtering purposes.

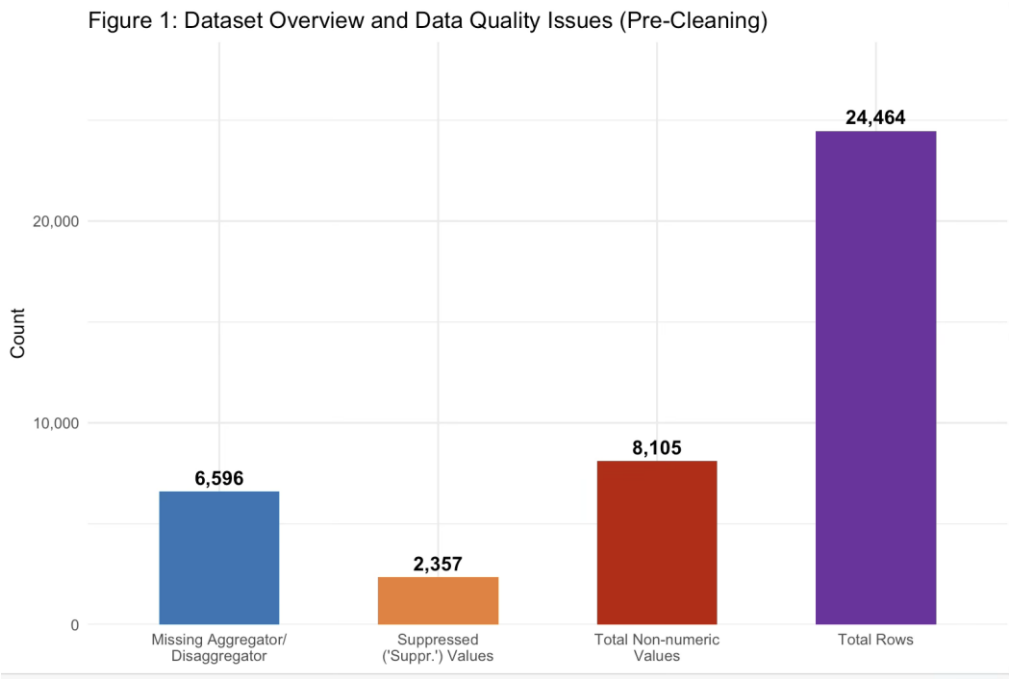

Figure 1: Dataset Overview and Data Quality Issues (Pre-Cleaning)

2. Data Analysis:

2.1 Dataset Overview — Descriptive Statistics

Table 2 summarizes the key characteristics and descriptive statistics of the cleaned dataset.

Table 2: Dataset Overview and Key Descriptive Statistics

Characteristic	Value
Total rows	24,464
Total columns	11 (original) + 3 derived after cleaning = 14
Substances	2 — Opioids and Stimulants
Harm sources	4 — Deaths, Hospitalizations, ED Visits, EMS Responses
Temporal coverage	January 2016 – September 2025
Time granularities	Annual (21,036 rows) and quarterly (3,428 rows)
Geographic regions	19 (national, 13 provinces/territories, 5 sub-regions)
Suppressed values	2,357 rows (9.6%) — privacy protection for small counts

Missing Aggregator/Disaggregator	6,596 rows (27.0%) — by design for overall totals
Mean annual national opioid deaths (2016-2024)	5,652
Peak opioid deaths	8,020 (2023)
Minimum opioid deaths	2,832 (2016)

2.2 National Opioid Death Trend, 2016–2024

Table 3 and Figure 2 present the national count and crude rate of apparent opioid toxicity deaths from 2016 to 2024.

Table 3: National Apparent Opioid Toxicity Deaths, 2016–2024

Year	Deaths (n)	Crude Rate per 100K	YoY Change
2016	2,832	7.8	—
2017	3,928	10.7	+38.7%
2018	4,253	11.5	+8.3%
2019	3,585	9.5	-15.7%
2020	6,166	16.2	+72.0%
2021	7,546	19.7	+22.4%
2022	7,227	18.6	-4.2%
2023	**8,020**	20.0	+11.0%
2024	7,313	17.7	-8.8%

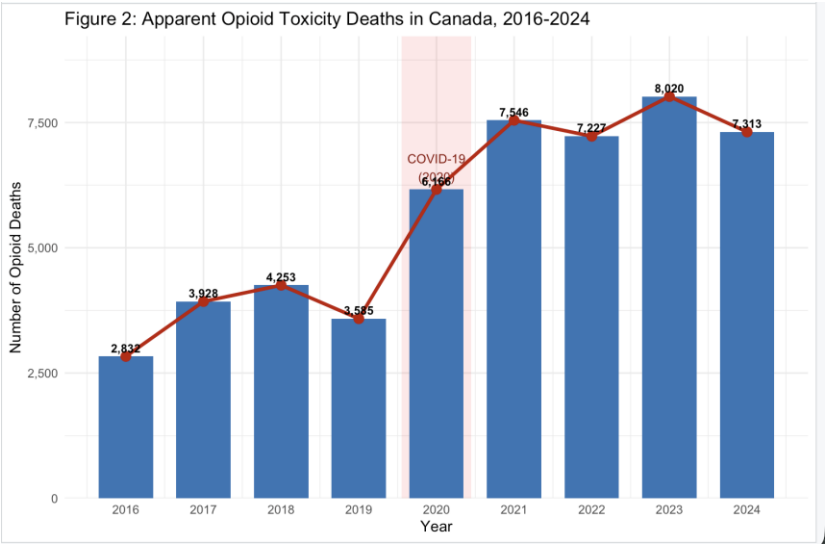

Figure 2: Apparent Opioid Toxicity Deaths in Canada, 2016-2024

The trend reveals a crisis that accelerated in two distinct waves. The first wave (2016–2018) saw deaths increase from 2,832 to 4,253, driven by the emergence of illicit fentanyl in the drug supply. A brief decline in 2019 (3,585 deaths) was followed by a catastrophic second wave

beginning in 2020, when deaths surged by 72% in a single year — from 3,585 to 6,166. Deaths peaked in 2023 at 8,020 — nearly three times the 2016 level — before declining modestly to 7,313 in 2024. Despite this recent decline, the 2024 figure remains more than 2.5 times higher than the pre-pandemic 2016 baseline, indicating the crisis remains severe.

2.3 Geographic Distribution — Deaths by Province, 2024

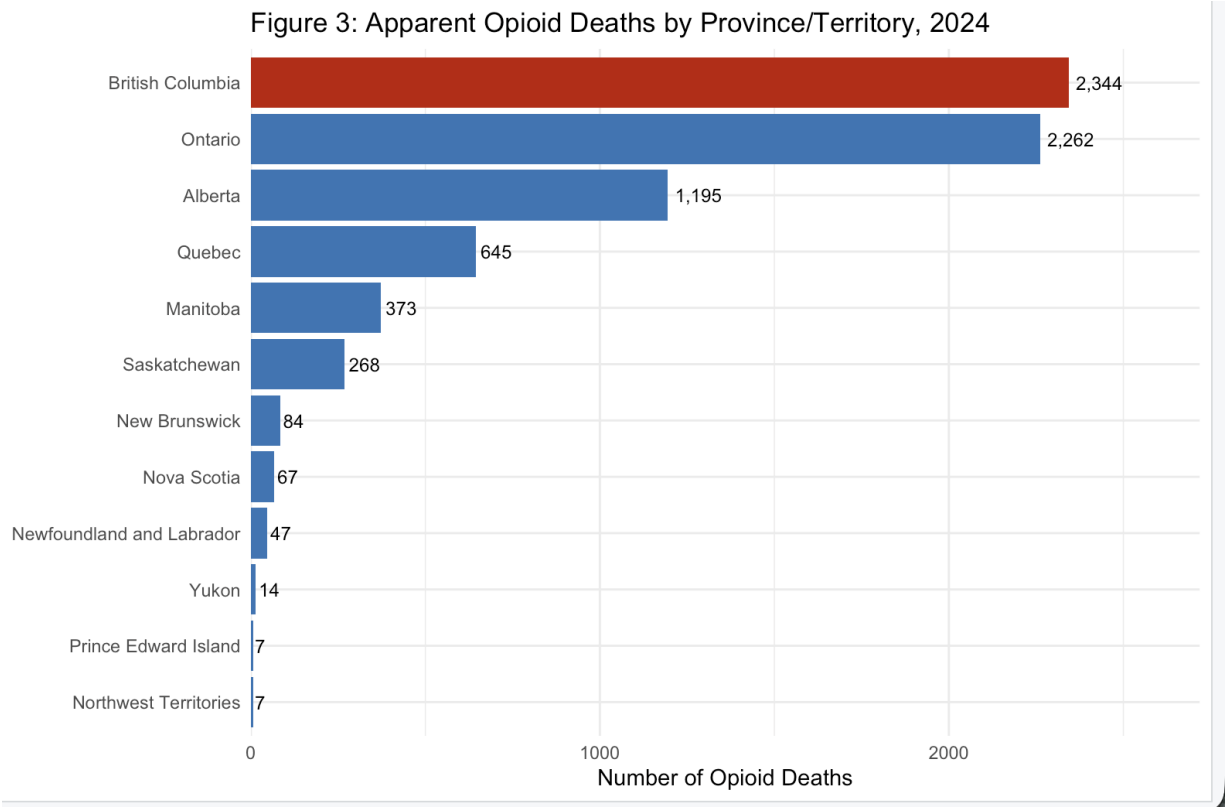

Figure 3: Apparent Opioid Deaths by Province/Territory, 2024

British Columbia leads all provinces with 2,344 deaths in 2024, narrowly ahead of Ontario with 2,262. Together, these two provinces account for 63% of all opioid deaths despite representing 52% of Canada's population. Alberta (1,195) ranks third. BC declared a public health emergency over illicit drug toxicity in April 2016 — the first Canadian province to do so — and has maintained that status continuously. The territories report very low absolute counts due to smaller populations, but their crude rates per 100,000 can be disproportionately high.

2.4 Opioids vs. Stimulants

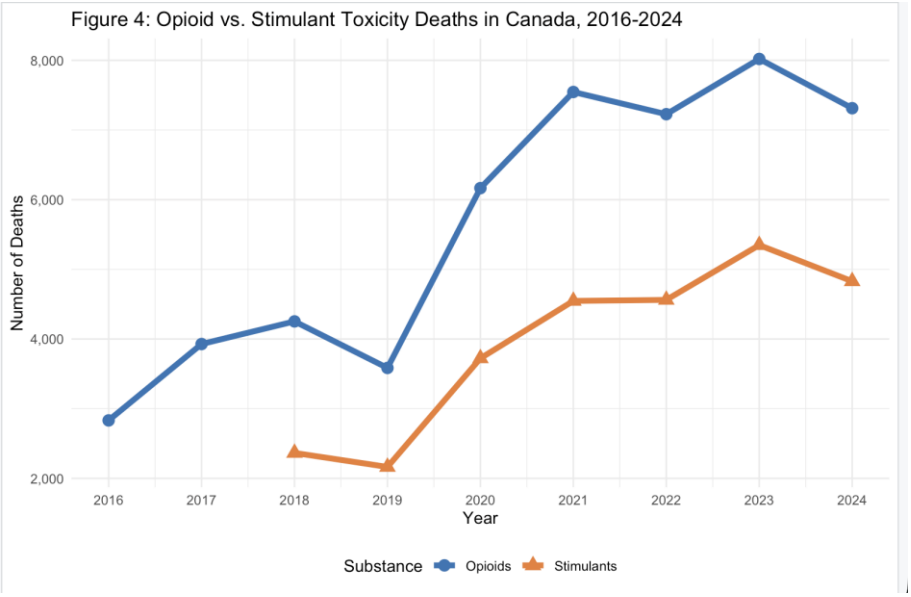

Figure 4: Opioid vs. Stimulant Toxicity Deaths in Canada, 2016-2024

While opioids dominate in absolute death counts, stimulant-related deaths have also grown significantly over the same period. Stimulant deaths rose from 2,164 in 2019 to 5,348 in 2023. These two categories are not mutually exclusive — many deaths involve both opioids and stimulants simultaneously, reflecting the increasingly complex nature of the illicit drug supply. The parallel rise of both curves confirms this is a poly-drug crisis, not simply an opioid crisis.

2.5 Deaths by Age Group

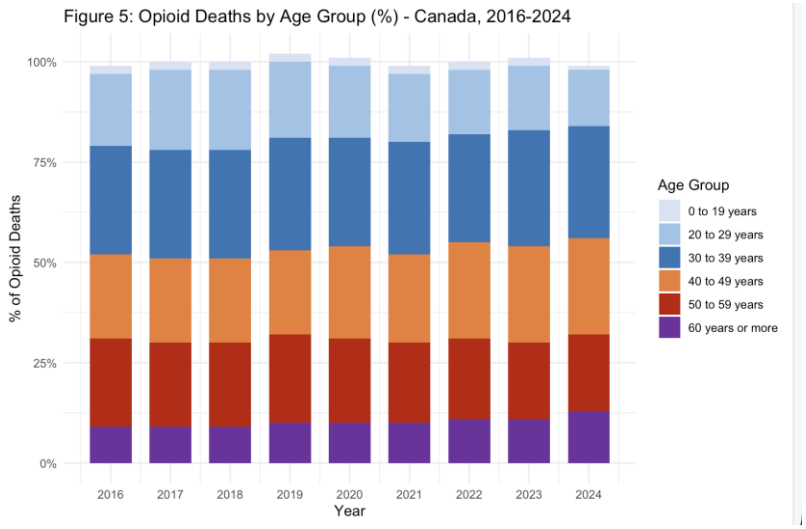

Figure 5: Opioid Deaths by Age Group (%) — Canada, 2016-2024

Adults aged 30 to 39 years consistently account for the largest share of opioid deaths — 28% in 2024 — a pattern that has remained stable since 2016. The 40 to 49 year group (24%) and 50 to 59 year group (19%) together account for an additional 43%, meaning working-age adults

between 30 and 59 represent approximately 71% of all opioid deaths. The 60 years or more group has grown from 9% in 2016 to 13% in 2024. The stability of this age distribution over nine years suggests the crisis is killing more people across the same age groups at a higher rate, rather than shifting in its demographic reach.

2.6 Deaths by Sex

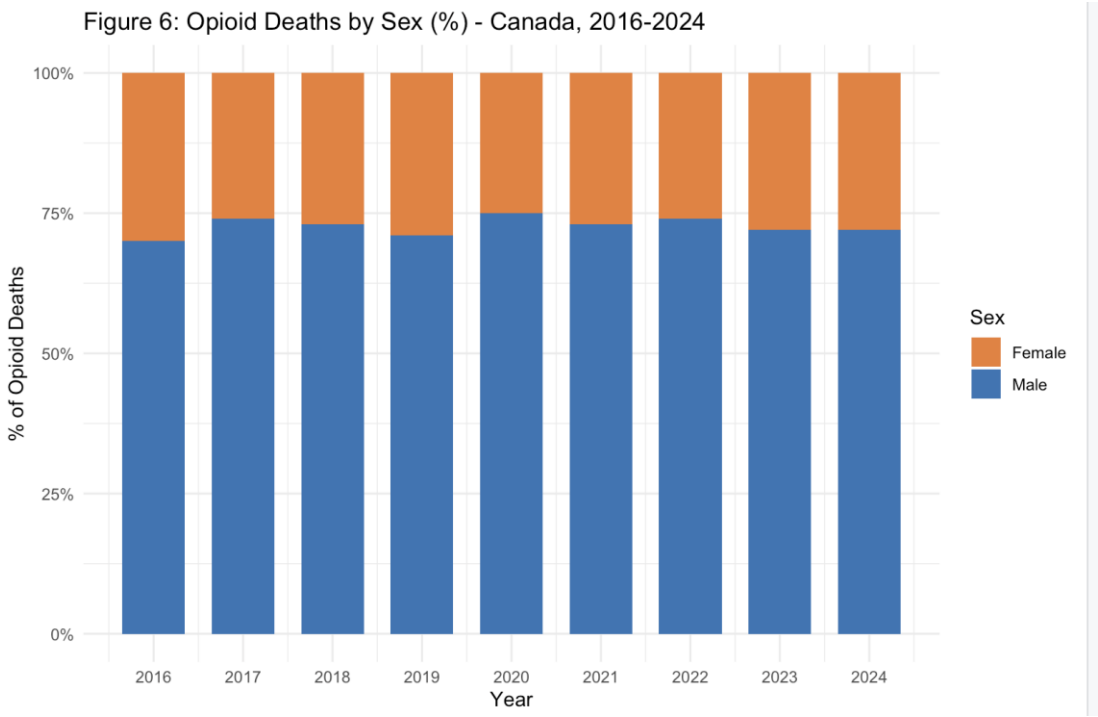

Figure 6: Opioid Deaths by Sex (%) — Canada, 2016-2024

Males account for approximately 72–75% of all opioid deaths every year from 2016 to 2024 — a disparity that has remained virtually unchanged throughout the entire surveillance period. In 2024, males represented 72% of deaths. This consistent gender gap reflects higher rates of illicit drug use among men, greater risk-taking behaviour, and lower uptake of harm reduction services. The stability of this pattern across nine years makes it one of the most structurally consistent findings in the entire dataset.

2.7 Fentanyl Involvement

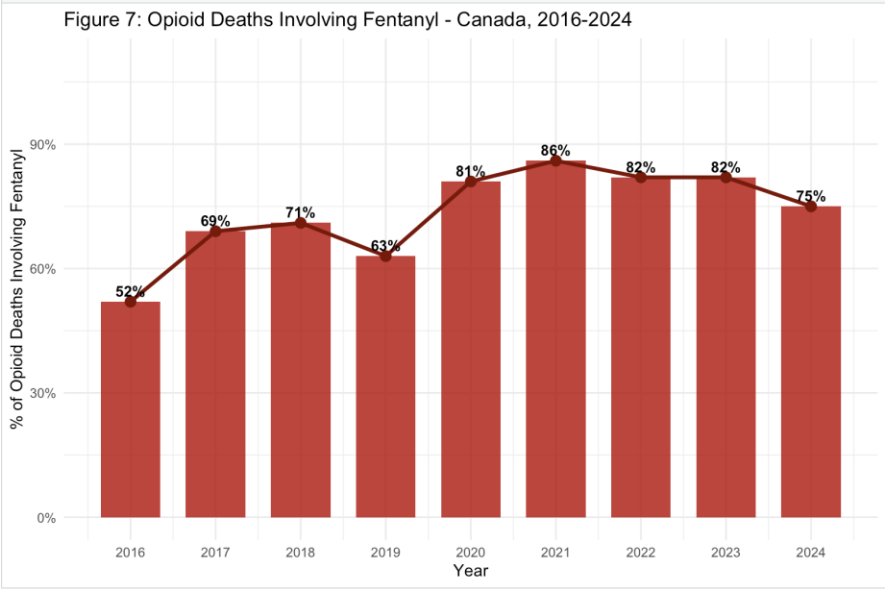

Figure 7: Opioid Deaths Involving Fentanyl — Canada, 2016-2024

Fentanyl was involved in 52% of opioid deaths in 2016. By 2021, that proportion had climbed to 86% — meaning that in nearly nine out of ten opioid deaths, fentanyl or an analogue was detected. The 2024 figure is 75%, reflecting some decline from the 2021 peak but still representing a dominant role for this single synthetic opioid. Fentanyl is estimated to be 50 to 100 times more potent than morphine. This shift from 52% to 75% over eight years illustrates how completely fentanyl has transformed Canada's illicit drug supply.

2.8 All Four Harm Types

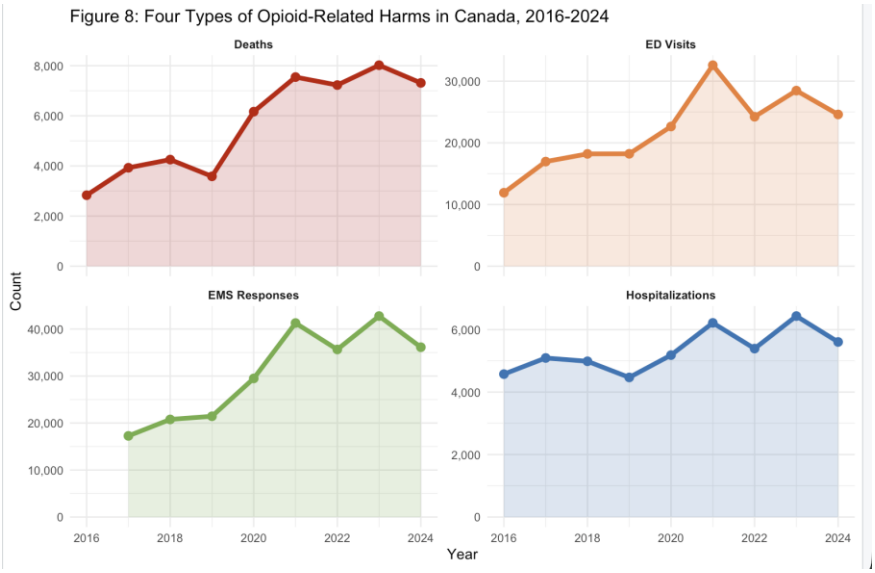

Figure 8: Four Types of Opioid-Related Harms in Canada, 2016-2024

All four opioid-related harm types — deaths, hospitalizations, ED visits, and EMS responses — show concordant trends: rising from 2016, a sharp acceleration after 2020, and a tentative

plateau or modest decline after 2022-2023. This parallel behaviour confirms that the observed trends in deaths are not artifacts of changing mortality coding practices, but are corroborated by independent data streams. Descriptive statistics across harm sources: Deaths (mean 5,652; range 2,832–8,020), Hospitalizations (mean 5,328; range 4,469–6,433), ED Visits (mean 21,984; range 11,912–32,592), EMS Responses (mean 30,599; range 17,269–42,756).

2.9 Geographic Heatmap — Crude Death Rates

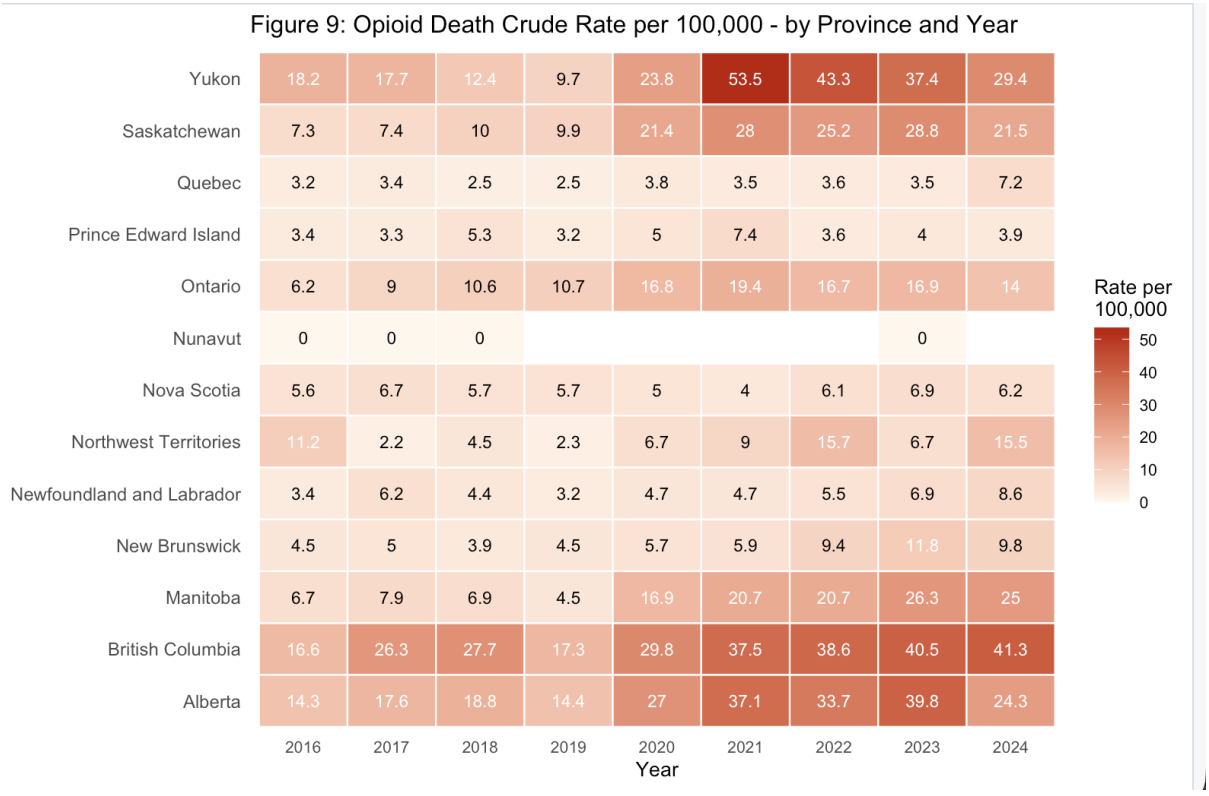

Figure 9: Opioid Death Crude Rate per 100,000 by Province and Year

The heatmap provides the most comprehensive geographic-temporal view. British Columbia has the highest crude death rates of any province in virtually every year, reaching 41.3 per 100,000 in 2024. Yukon recorded the single highest rate of any jurisdiction — 53.5 per 100,000 in 2021. Alberta's rate roughly tripled from 2016 to 2023. Ontario, despite having the second highest absolute death count, has a substantially lower rate per capita due to its much larger population (14.0 per 100,000 in 2024). Quebec consistently shows the lowest rates across all years. Empty cells represent suppressed data where counts fell below the privacy threshold.

2.10 Distribution Analysis — Histogram

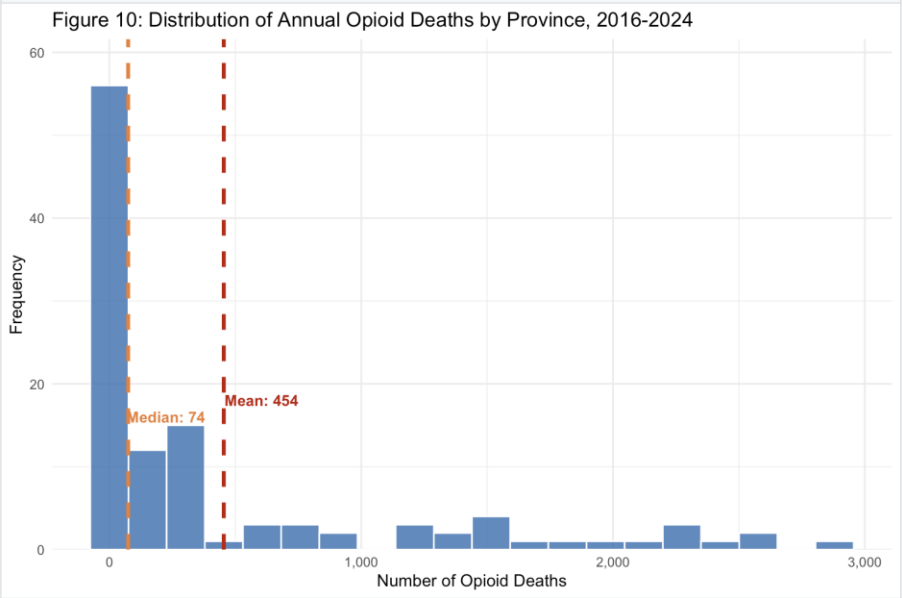

Figure 10: Distribution of Annual Opioid Deaths by Province, 2016-2024

The histogram reveals a strongly right-skewed distribution. The majority of province-year observations (over 55 cases) fall in the lowest bin near zero, representing small provinces and territories with low annual death counts. The mean (454) is far higher than the median (74.5), confirming the presence of extreme high-value outliers — specifically British Columbia and Ontario — that pull the average upward. This skewness is an important characteristic to note: the national "average" provincial death count is not representative of the typical province, which actually experiences far fewer deaths.

2.11 Variability Analysis — Boxplot

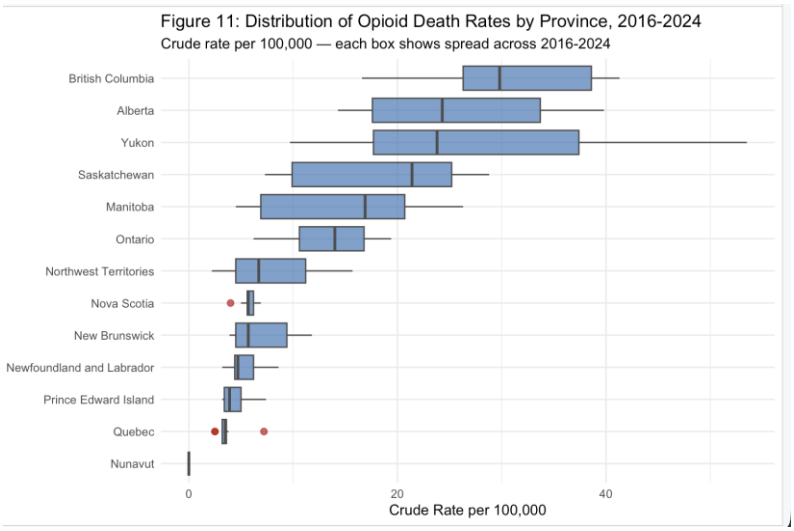

Figure 11: Distribution of Opioid Death Rates by Province, 2016-2024

The boxplot shows the spread of crude death rates across years for each province. British Columbia has both the highest median rate and one of the widest boxes, indicating consistently

high rates with substantial year-to-year variation. Yukon shows the longest whisker extending beyond 50 per 100,000 — the outlier year of 2021 identified in the heatmap. Quebec and Nunavut have very compact boxes close to zero, reflecting consistently low rates. Red outlier dots on Nova Scotia and Quebec represent individual years that deviated significantly from those provinces' typical range. This variability confirms that no two provinces experienced this crisis identically.

2.12 Relationship Analysis — Scatterplot

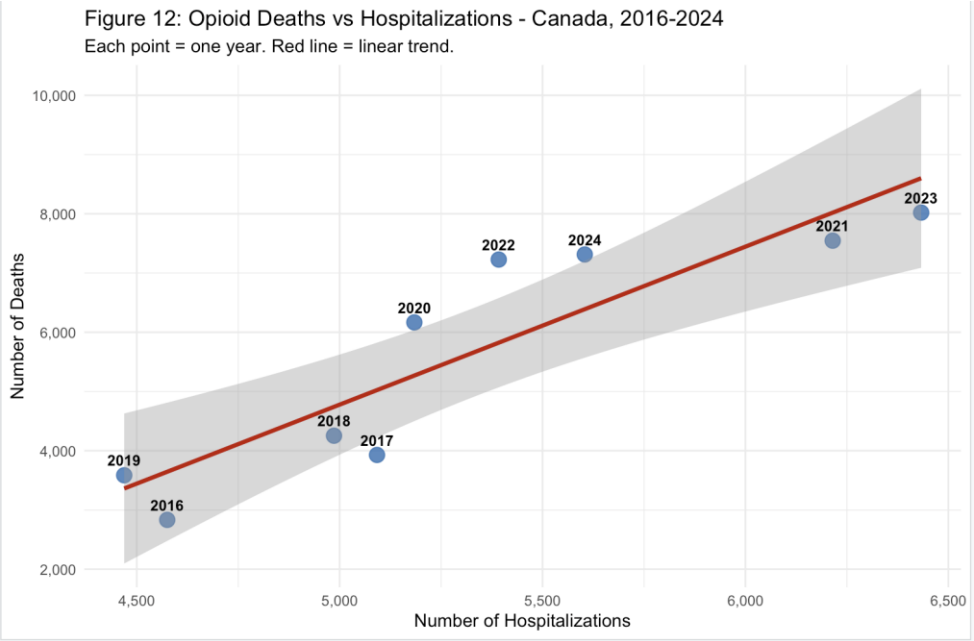

Figure 12: Opioid Deaths vs. Hospitalizations — Canada, 2016-2024

The scatterplot reveals a strong positive linear relationship between annual opioid deaths and hospitalizations nationally. As hospitalizations increase, deaths increase proportionally. The red trend line (linear regression) fits the data well, with most years clustering closely around it. This correlation suggests that hospitalizations serve as a reliable leading indicator of mortality burden. One notable observation is that 2020 appears slightly below the trend line — indicating that despite a large spike in deaths, hospitalizations did not increase as dramatically, possibly because many COVID-era overdoses occurred in isolation with no hospitalization before death.

3. Summary

This exploratory data analysis of the PHAC Opioid- and Stimulant-related Harms in Canada dataset (2016–2025) reveals a national drug toxicity crisis of extraordinary severity and growing complexity. The following are the key takeaways:

- National opioid deaths nearly tripled between 2016 (2,832) and the 2023 peak (8,020), with a 72% single-year surge in 2020 attributable to the compounding effects of COVID-19 on drug use patterns and harm reduction access.
- British Columbia and Ontario together account for 63% of national opioid deaths in 2024, though BC's crude rate per capita (41.3 per 100,000) is dramatically higher than any other province.
- Fentanyl's role grew from 52% of opioid deaths in 2016 to a peak of 86% in 2021, and remains at 75% in 2024 — cementing synthetic opioids as the primary driver of Canada's overdose mortality.
- Males consistently account for approximately 72% of opioid deaths, and adults aged 30 to 39 are the most affected age group — both patterns completely stable across all nine years of surveillance.
- All four harm types (deaths, hospitalizations, ED visits, EMS responses) show concordant trends, validating that the pattern reflects real-world harm escalation rather than data artifacts.
- The distribution of provincial death counts is strongly right-skewed — most provinces have low counts, but BC and Ontario are extreme outliers that dominate the national picture.
- 9.6% of data values are suppressed for privacy, predominantly in small provinces and territories — a limitation that likely leads to underestimation of the true burden in remote communities.

Questions for further investigation:

- Can the quarterly data be used to detect seasonal patterns in overdose risk, and do these patterns differ by province?
- What explains the divergence between BC's persistently high crude rates and the rest of Canada, even after controlling for population size and drug supply differences?
- Can this dataset be linked to socioeconomic indicators at the Census Subdivision level to identify community-level risk factors?
- As 2025 full-year data becomes available, will the recent decline in deaths continue, or was 2024 a temporary dip?
- Does the strong correlation between deaths and hospitalizations hold at the provincial level, or only nationally?

The PHAC dataset is one of the most policy-relevant and analytically rich public health surveillance datasets currently available in Canada. Its quarterly update cadence, multi-dimensional demographic breakdowns, and national geographic coverage make it an excellent foundation for both descriptive analysis and predictive modelling of substance-use harms. Future analytical work could apply regression modelling to forecast deaths by province, or cluster analysis to identify provinces with similar harm trajectories but divergent policy responses.

References

Public Health Agency of Canada. (2026, March). Opioid- and Stimulant-related Harms in Canada [Dataset]. Public Health Infobase. https://health-infobase.canada.ca/substance-related-harms/opioids-stimulants/